



Connect a GitHub Repo with AWS



yakubu abdullahi yusuf

```
ec2-user@ip-172-31-72-4 ~]$ sudo dnf update -y
sudo dnf install git -y
last metadata expiration check: 22:29:24 ago on Tue Nov 25 23:16:11 2025.
dependencies resolved.
nothing to do.
[complete]
last metadata expiration check: 22:29:25 ago on Tue Nov 25 23:16:11 2025.
dependencies resolved.
```

Package	Architecture	Version	Repository	Size
installing:				
git	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	53 k
installing dependencies:				
git-core	x86_64	2.50.1-1.amzn2023.0.1	amazonlinux	4.9 M
git-core-doc	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	2.8 M
perl-error	noarch	1:0.17029-5.amzn2023.0.2	amazonlinux	41 k
perl-file-find	noarch	1.37-477.amzn2023.0.7	amazonlinux	25 k
perl-git	noarch	2.50.1-1.amzn2023.0.1	amazonlinux	41 k
perl-termreadkey	x86_64	2.30-9.amzn2023.0.2	amazonlinux	36 k
perl-lib	x86_64	0.65-477.amzn2023.0.7	amazonlinux	15 k

```
Transaction Summary
Install 8 Packages

Total download size: 7.9 M
Installed size: 41 M
Downloading Packages:
1/8: git-2.50.1-1.amzn2023.0.1.x86_64.rpm           1.6 MB/s | 53 kB  00:00
2/8: git-core-doc-2.50.1-1.amzn2023.0.1.noarch.rpm  43 MB/s | 2.8 MB 00:00
3/8: perl-error-0.17029-5.amzn2023.0.2.noarch.rpm   1.1 MB/s | 41 kB  00:00
4/8: git-core-2.50.1-1.amzn2023.0.1.x86_64.rpm      47 MB/s | 4.9 MB 00:00
5/8: perl-file-find-1.37-477.amzn2023.0.7.noarch.rpm 617 kB/s | 25 kB 00:00
6/8: perl-termreadkey-2.30-9.amzn2023.0.2.x86_64.rpm 1.4 MB/s | 36 kB 00:00
7/8: perl-lib-0.65-477.amzn2023.0.7.x86_64.rpm      1.4 MB/s | 15 kB 00:00
8/8: git-2.50.1-1.amzn2023.0.1.x86_64.rpm           1.6 MB/s | 53 kB 00:00
```



Introducing Today's Project!

This is the Day 2 of the DevOps 7 Days Challenge. In this project, I will be Connecting a GitHub Repo with AWS. I will be connecting my web app project to a GitHub repo, then try making changes to my web app code and see the updates in Github.

Key tools and concepts

Services I used were Git, GitHub, Vscode, and ec2 instance Key concepts I learnt include Git and GitHub, staging, commits, push and generating tokens for secure logins.

Project reflection

This project took me approximately one hour, It was very fun and interesting project.

I did this project because I want to learn more about CICD.

This project is part two of a series of DevOps projects where I'm building a CI/CD pipeline! I'll be working on the next project tomorrow.



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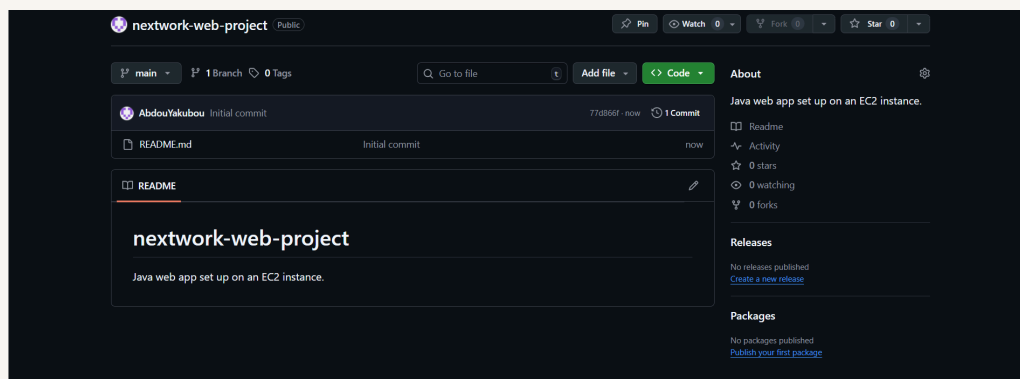
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Git and GitHub

Git is a version control system, Git helps track every change I make, and lets me go back to any version of my work. I installed Git using the commands "sudo dnf install git -y"

GitHub is a storage space for the projects engineers work on. I will be using GitHub to have a more user-friendly way of seeing my code changes and a more friendly way to use Git.





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My local repository

Git repositories are collection of folders that contains all files, code, and change in history of a project and all these are kept using the Git version control system.

Git init is a command that initializes a new, empty Git repository in the current working directory. It tells Git to start tracking the files in your project folder. I ran git init in my Weeb app folder on my ec2 instance.

A branch in Git is like a separate path of work in your project. It allows me to make changes, test new ideas, or build new features without touching the main project, which is also known as master branch.



```
/home/ec2-user/nextwork-web-project
[ec2-user@ip-172-31-72-4 nextwork-web-project]$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/ec2-user/nextwork-web-project/.git/
```



To push local changes to GitHub, I ran three commands

git add

The first command I ran was "git add ." A staging area is a temporary space where Git keeps track of changes you want to include in your next commit. This keeps all your changes in a waiting space for final approval before commit.

git commit

The second command I ran was "git commit -m" Using '-m' is a flag which allows you to tag a message along with your commit. It leaves messages that describes what the commit is about.

git push

The third command I ran was "git push -u origin master", which means pushing my master to the remote repository, which sets up a connection upstream. Using '-u' means that Git should remember here to push and pull changes from.



Authentication

When I commit changes to GitHub, Git asks for my credentials because Git needs to double check that you have the right to push any changes to the remote origin your local repo is connected with.

Local Git identity

Git needs my name and email because Git needs author information for commits to track who made what change.

Running git log showed me the commits ive made together with the author's name.

```
[ec2-user@ip-172-31-72-4 nextwork-web-project]$ git log
commit 135afa4127aa7b33c712bc39888f8a6f6e4a73dc (HEAD -> master, origin/master)
Author: EC2 Default User <ec2-user@ip-172-31-72-4.ec2.internal>
Date:   Wed Nov 26 23:32:52 2025 +0000

    Updated index.jsp with new content
[ec2-user@ip-172-31-72-4 nextwork-web-project]$
```



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GitHub tokens

GitHub authentication failed when I entered my password because password can be intercepted over the network.

A GitHub token is a secure key that acts like a password. I'm using one in this project because GitHub no longer allows passwords for Git operations over HTTPS.

I could set up a GitHub token by going to the settings options in GitHub, then going into the developer options tab, then generate new token (classic), and I also check the repo scopes.



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GitHub Apps

OAuth Apps

Personal access tokens

Fine-grained tokens

Tokens (classic)

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of a password for Git over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

Generated for EC2 Instance Access. This is a part of NextWork's 7 D.

What's this token for?

Expiration

7 days (Dec 04, 2025)

The token will expire on the selected date

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repostatus	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo_invite	Access repository invitations
☒ security_events	Read and write security events



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Making changes again

I wanted to see Git working in action, so I edited my index.jsp file but I couldn't see the changes in my GitHub repo initially because I didn't use Git to stage, commit and push the changes to GitHub repo.

I finally saw the changes in my GitHub repo after staging, committing and pushing the change using the git commands.

The screenshot shows a GitHub commit interface for a file named `index.jsp`. The commit message is "Added new line to index.jsp" by user "yakubu". The commit hash is "9ba1083" and it was made "3 minutes ago". The file statistics show "6 lines (6 loc) · 151 Bytes". The code is displayed in a dark-themed editor with the following content:

```
1 <html>
2 <body>
3 <h2>Hello YEBI</h2>
4 <p>If you see this line in GitHub, that means your changes are getting pushed to the cloud repo</p>
5 </body>
6 </html>
```



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